

Lab 24 Worksheet — Storage and RAID Fault Diagnosis with S.M.A.R.T.

Name: ____ Date: ____

Exam objective: Diagnose storage faults using S.M.A.R.T. data, symptom analysis and RAID status, and choose the correct recovery action for each RAID level (Core 1 objective 5.3).

Record as you go

Step	What you did	What you observed
1	Record the first rule of storage troubleshooting: if the data matters and the drive is failing, back it up before you do anything else. Diagnostics can be the final straw for a dying drive.	
2	Check drive health on Windows using the built-in interface and record the status reported for each physical disk.	
3	Open Disk Management and record each volume's file system, capacity, free space and status.	
4	Open the Killercodea Ubuntu playground and install the S.M.A.R.T. monitoring tools to read raw drive attributes.	
5	List the block devices available so you know which device names to query.	
6	Read the S.M.A.R.T. attributes for a device and record the overall health self-assessment result.	
7	Record the S.M.A.R.T. attributes that matter most: reallocated sector count, current pending sector count, offline uncorrectable and read error rate — any non-zero value on the first three is a replace signal.	
8	Record the correct response to a S.M.A.R.T.	

Step	What you did	What you observed
	warning: back up immediately and replace the drive. S.M.A.R.T. reports a drive that is already failing, not one that might fail eventually.	
9	Map the audible and behavioural symptoms: clicking or grinding means imminent mechanical failure, and extended read and write times or falling IOPS mean the drive is retrying failing sectors.	
10	Map 'bootable device not found' to its three causes: wrong boot order in firmware, a dead drive, or a corrupt boot record — and give the test that distinguishes them.	
11	Build the RAID recovery table: RAID 1, 5 and 10 survive one disk failure and rebuild after replacement, while RAID 0 has no redundancy so any failure means a full restore from backup.	
12	Record the RAID rebuild warning: a rebuild puts every remaining disk under sustained full load, which is exactly when a second ageing disk in the same batch tends to fail — so verify the backup before starting one.	

Verification

Success criterion: Your record names at least four S.M.A.R.T. attributes and states the correct action for a non-zero reallocated sector count; the recovery table gives a distinct action for all four RAID levels; and the rebuild risk is stated explicitly.

- I completed every step in the lab.
- My result meets the success criterion above.
- I recorded my evidence (screenshots, output, completed tables).

Reflection

What surprised you in this lab?

Where would you apply this on the job?

What do you still need to revise before the exam?
